

3D-Printed Prosthesis with an Articular Interface for Anatomical Acetabular Reconstruction After Type I + II (+ III) Internal Hemipelvectomy

Clinical Outcomes and Finite Element Analysis

Xin Huang, MD*, Donghua Huang, MD*, Nong Lin, MD, Xiaobo Yan, MD, Hao Qu, MD, and Zhaoming Ye, PhD

Investigation performed at the Musculoskeletal Tumor Center, Department of Orthopedics, The Second Affiliated Hospital of Zhejiang University School of Medicine, Hangzhou, Zhejiang, People's Republic of China

Background: Pelvic reconstruction after type I + II (or type I + II + III) internal hemipelvectomy with extensive ilium removal is a great challenge. In an attempt to anatomically reconstruct the hip rotation center (HRC) and achieve a low mechanical failure rate, a custom-made, 3D-printed prosthesis with a porous articular interface was developed. The aim of this study was to investigate the clinical outcomes of patients treated with this prosthesis.

Methods: This retrospective cohort study included 28 patients with type I + II (+ III) internal hemipelvectomy through the articular interface of the sacroiliac joint and managed with a prosthesis at a single center between August 2016 and August 2021. Complications and oncological outcomes were analyzed. The position of the reconstructed HRC was assessed and lower-limb function was evaluated. Biomechanical analyses of different fixation modes of the prosthesis were conducted using finite element analysis.

Results: The displacement distance of the HRC from preoperatively to postoperatively was a mean (and standard deviation) of 14.12 ± 8.75 mm. The rate of implant-related complications was 14.3% (4 of 28) for prosthetic breakage, 14.3% (4 of 28) for aseptic loosening, 7.1% (2 of 28) for dislocation, and 7.1% (2 of 28) for deep infection. The mean Musculoskeletal Tumor Society (MSTS)-93 score was 18.2. The aseptic loosening rate was significantly greater for prostheses fixed with 3 sacral screws (4 of 10, 40.0%) than for those fixed with 4 (0 of 10, 0%) or 5 screws (0 of 8, 0%) ($p = 0.024$). The prosthetic breakage rate was lower in patients who underwent lumbosacral fixation (0 of 13, 0%) than in those who did not (4 of 15, 26.7%), although the difference did not reach significance ($p = 0.102$). Biomechanical analyses suggested that the addition of lumbosacral fixation or increasing the number of sacral screws from 3 to 4 or 5 visibly reduced the peak stress of the sacral screws.

Conclusions: The use of a 3D-printed prosthesis with an articular interface for pelvic reconstruction demonstrated stable prosthetic fixation, anatomical acetabular reconstruction, and acceptable early functional outcomes.

Level of Evidence: Therapeutic Level III. See Instructions for Authors for a complete description of levels of evidence.

Restoring patients' ability to stand and walk after pelvic tumor resection is challenging¹. A variety of prostheses, such as a stemmed acetabular pedestal cup² and cus-

tom-made³⁻⁶ and modular hemipelvic prostheses⁷⁻¹⁰, have been developed to achieve better outcomes of pelvic reconstruction after the resection of periacetabular tumors. The residual ilium

*Xin Huang, MD, and Donghua Huang, MD, contributed equally to this work.

Disclosure: This work was funded by the Zhejiang Provincial Natural Science Foundation of China (LY23H060010, LQ21H060004) and the National Natural Science Foundation of China (82372382, 82002333). These funding sources did not play a role in the investigation. The **Disclosure of Potential Conflicts of Interest** forms are provided with the online version of the article (<http://links.lww.com/JBJS/1229>).

Copyright © 2024 The Authors. Published by The Journal of Bone and Joint Surgery, Incorporated. This is an open access article distributed under the terms of the [Creative Commons Attribution-Non Commercial-No Derivatives License 4.0](https://creativecommons.org/licenses/by-nc-nd/4.0/) (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

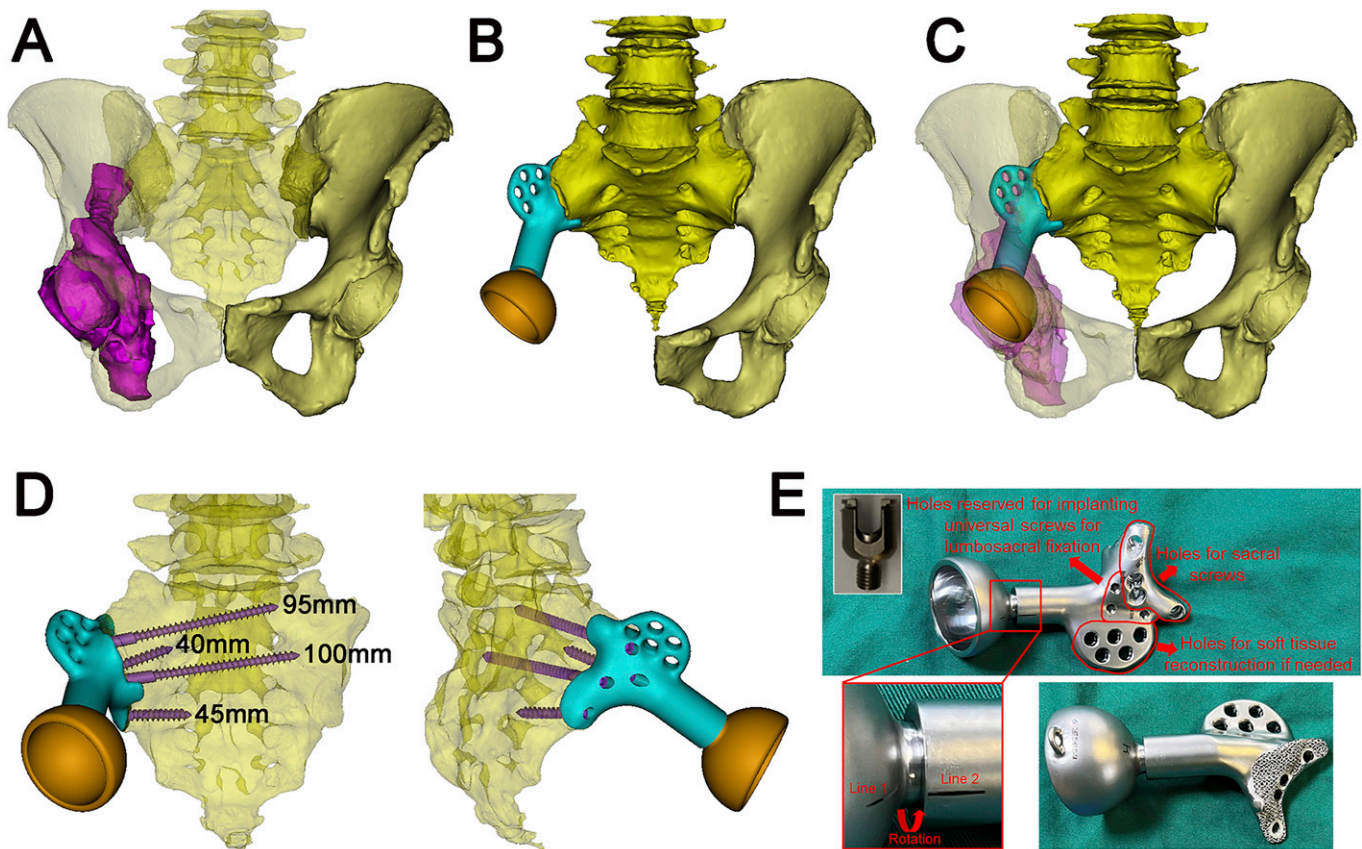


Fig. 1
Preoperative simulation and prosthesis design and fabrication. The pelvic model (yellow), iliac component (cyan), acetabular component (khaki), sacral screw (purple), and tumor (purplish red) are shown. **Fig. 1-A** First, preoperative thin-slice computed tomography (CT) and magnetic resonance (MR) images were imported into Mimics V20.0 software (Materialise) to reconstruct virtual 3D pelvic models. **Fig. 1-B** Second, an individualized prosthesis was designed and optimized. The porous articular interface of the prosthesis was designed to seamlessly fit the articular interface of the sacrum. **Fig. 1-C** After the above procedures, the location of the acetabular cup was determined according to the location of the native hip joint. The inclination and anteversion of the acetabular cup were artificially set to 40° and 15°, respectively. **Fig. 1-D** On the basis of the sacral geometry in the patient, 3 to 5 sacral screws were designed to fix the prosthesis to the sacrum. The trajectories and lengths of the sacral screws were strictly planned to avoid penetrating into the sacral canal or anterior neurovascular plexus. A sacral tray was introduced to fully fit the ventral sacral interface. **Fig. 1-E** Third, a metal prosthesis with a porous articular interface was 3D-printed and fabricated. There was a line on the surface of the cup and another on the base (lines 1 and 2). If the 2 lines were aligned, the inclination and anteversion of the acetabular cup would reach the preset angles (40° and 15°, respectively). During prosthetic fixation, the acetabular cup could be rotated to change the anteversion angle according to the intraoperative situation. Usually, a total of approximately 10 days is needed for the preparation of the prosthesis, including 1 to 2 days for the prosthesis design, 5 to 6 days for the 3D printing and further fabrication of the prosthesis, and 2 to 3 days for prosthesis transportation and disinfection measures.

can provide reliable osseous anchorage for prostheses, which is conducive to mechanical fixation. However, most of these methods may not be suitable for type I + II (or type I + II + III) reconstruction¹¹ when both the ilium and acetabulum are extensively excised.

In the present study, a newly designed, 3D-printed, custom-made hemipelvic prosthesis (Chunli) (Fig. 1) with a porous articular interface was developed for pelvic reconstruction after type I + II (+ III) internal hemipelvectomy. To achieve precise matching between the articular interface of the prosthesis and the sacrum, we modified the osteotomy method. Instead of using a traditional line saw for direct bone cutting, we employed an electric knife to gradually

separate the sacroiliac joint, ensuring complete preservation of the sacral articular interface. Benefiting from a seamless “bone-implant” interface, the position of the prosthesis could be uniquely determined during installation allowing placement of the acetabular cup in the native position, which could maintain the function of the affected limb well and decrease prosthesis-related complications. More importantly, our method allowed the use of a custom-made prosthesis without requiring navigation or custom cutting guides.

The specific goals of this study were to (1) evaluate whether anatomical reconstruction of the acetabulum could be achieved with the new prosthesis in cases of type I + II (+ III) pelvic resection, (2) report the functional outcomes and complications

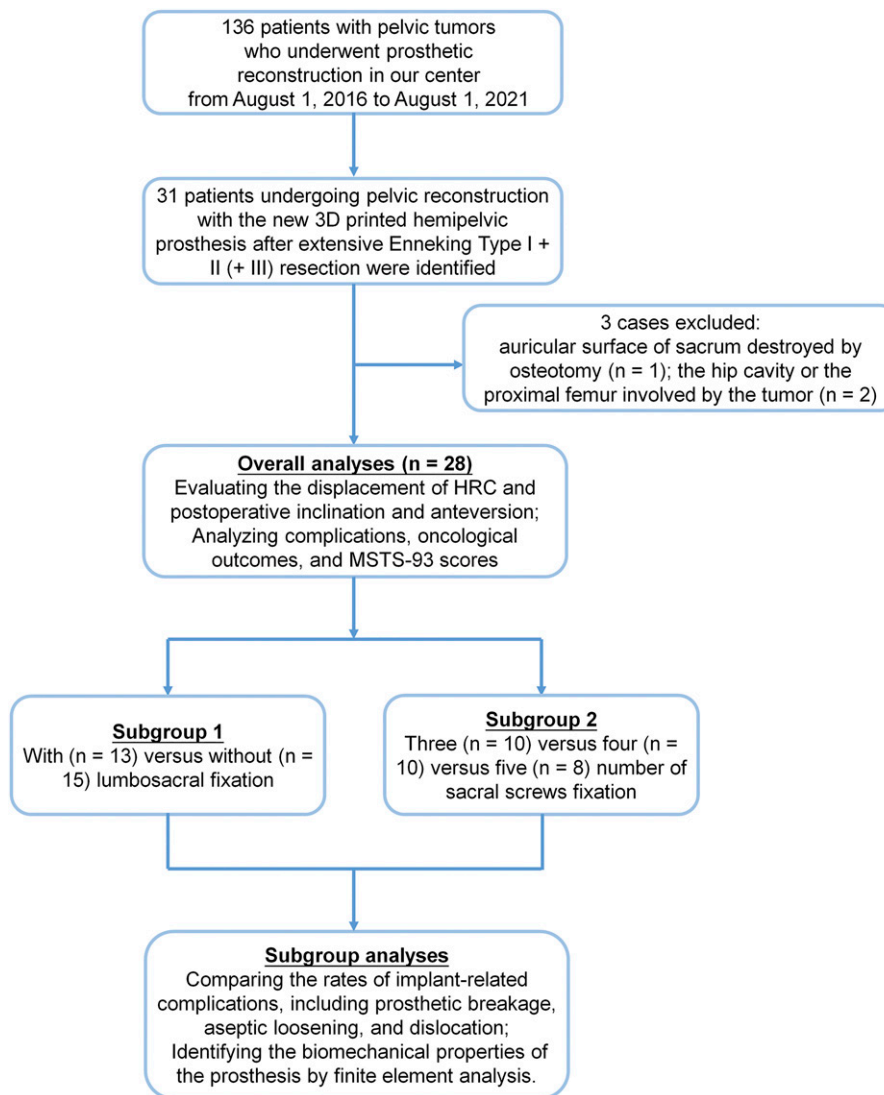


Fig. 2

A flow diagram depicting the processes of case inclusion and exclusion for the overall analyses and subgroup analyses. HRC = hip rotation center, MSTs = Musculoskeletal Tumor Society.

associated with this prosthesis at a minimum of 2 years of follow-up, and (3) identify an optimal design for prostheses with excellent biomechanical properties by finite element analysis.

Materials and Methods

Patients

The current study was approved by the institutional review board of our hospital (2021-0141). Between August 2016 and August 2021, a total of 136 patients underwent hemipelvectomy with prosthetic reconstruction for oncological indications in our musculoskeletal tumor center. The general indications for this prosthesis were periacetabular tumors with extensive iliac involvement, which resulted in a lack of remaining iliac bone for conventional hemipelvic prosthesis fixation after tumor resection. The contraindications were (1) invasion of the sacroiliac joint by the tumor, (2) a tumor

of the ilium with a huge soft-tissue mass hindering disarticulation of the sacroiliac joint, and (3) ankylosed sacroiliac joints. An extensive osteolytic lesion in the ilium near the sacroiliac joint was regarded as a relative contraindication. The process of inclusion and exclusion is shown in Figure 2. Overall, 28 patients were included in the study. The demographic and clinical characteristics are presented in Table I. Briefly, there were 22 patients with primary malignant tumors and 6 patients with metastatic disease. According to the Enneking and Dunham classification system¹¹, a type I + II resection was carried out for 12 patients, and a type I + II + III resection was performed for 16 patients. Three patients underwent pelvic radiographic imaging only, without computed tomography (CT) scans, for various reasons. Thus, 25 of the 28 patients were included in the assessment of the accuracy of implant installation (Table II).

TABLE I Demographics and Clinical Outcomes of Patients*

Patient	Sex	Age (yr)	Side	Diagnosis	Tumor Region ^{1,†}	Blood Loss (mL)	Operative Time (hr)	No. of Sacral Screws	Lumbosacral Fixation	Follow-up (mo)	Complications	Margin	Oncologic Outcome	Status	MSTS-93 Score [†]
1	M	69	R	CS	I + II + III	2,000	8.3	4	No	41.5		R0	DM†	AWD	20
2	F	65	L	CS	I + II	1,200	4.3	4	Yes	16.1		R0	LR, DM†	DOD	13
3	M	59	R	CS	I + II + III	1,000	7.5	4	Yes	22.3		R0		NED	24
4	M	62	L	RCCC	I + II	2,000	5.0	3	No	36.8	AL	R0	DM	DOD	10
5	F	44	R	UPS	I + II	1,400	7.3	3	Yes	34.0	SI	R1	LR, DM	DOD	9
6	M	55	R	RCCC	I + II	1,000	4.6	3	No	54.3	PB, AL	R0		DOD	13
7	M	27	R	CS	I + II + III	1,600	5.7	3	No	84.9	PB	R0		NED	28
8	M	56	R	PNET	I + II + III	2,000	5.3	4	No	38.7		R0	DM†	DOD	22
9	F	33	R	CS	I + II	1,600	7.5	4	No	39.2	PB	R0		NED	21
10	M	59	L	CS	I + II + III	1,600	7.6	3	No	68.1	PB, AL	R0		NED	13
11	M	32	L	CS	I + II + III	1,000	6.2	5	No	17.1	SI	R0	DM	DOD	22
12	M	49	L	CS	I + II + III	1,500	7.7	4	No	61.6	Dislocation, DI	R0		NED	14
13	F	50	L	OS	I + II + III	2,000	6.6	4	No	43.6		R0	DM†	DOD	20
14	F	61	L	UPS	I + II	1,500	5.2	3	Yes	40.6		R0		NED	14
15	F	61	L	OS	I + II	2,000	7.1	3	Yes	24.0		R0		DOD	15
16	M	60	R	RCCC	I + II	1,600	5.7	3	Yes	35.7	AL, SI	R0	DM†	AWD	16
17	M	43	R	OS	I + II	600	5.3	5	Yes	20.3		R0		DOD	15
18	F	70	L	FTC	I+II	1,500	5.8	3	Yes	32.4		R0		AWD	26
19	M	28	L	M-PEComa	I + II + III	2,000	7.1	5	Yes	30.5	DI	R0		NED	17
20	M	44	L	M-PEComa	I + II	1,000	4.7	5	Yes	26.3	Dislocation	R0		DOD	9
21	F	67	R	OS	I + II + III	500	3.8	5	No	28.9		R0		NED	24
22	F	56	L	LMS	I + II + III	2,000	6.1	4	No	28.2		R0	LR, DM	AWD	16
23	M	38	L	OS	I + II	1,000	5.5	4	Yes	24.0		R0		NED	21
24	M	58	L	CS	I + II + III	1,400	7.3	5	Yes	29.4		R0		NED	20
25	F	62	R	FTC	I + II + III	1,000	8.3	5	Yes	26.8		R0		AWD	22
26	M	66	R	CS	I + II + III	2,000	6.6	4	No	71.2		R0	DM†	AWD	23
27	M	38	R	CS	I + II + III	1,600	6.6	3	No	6.2		R0	DM	DOD	18
28	M	26	L	OS	I + II + III	800	4.7	5	No	31.7		R0		NED	25

*CS = chondrosarcoma, RCCC = renal clear cell carcinoma, UPS = undifferentiated pleomorphic sarcoma, PNET = primitive neuroectodermal tumors derived from esophagus, OS = osteosarcoma, FTC = follicular thyroid cancer, M-PEComa = malignant perivascular epithelioid cell tumor, LMS = leiomyosarcoma, I + II = iliac and periacetabular areas, I + II + III = iliac, periacetabular, and pubic areas, AL = aseptic loosening, SI = superficial infection, PB = prosthetic breakage, DI = deep infection, DM = distant metastasis, LR = local recurrence, R0 = negative surgical margin, R1 = positive surgical margin, NED = no evidence of disease, AWD = alive with disease, DOD = died of disease, and MSTS = Musculoskeletal Tumor Society. †Score at latest follow-up. ‡Patient experienced metastasis before surgery.

Prosthesis Design and Manufacture

The 3D-printed hemipelvic prosthesis was designed and manufactured in 3 main steps, with details shown in Figure 1.

Surgical Technique

After the tumor was routinely exposed, a distal osteotomy of the pubic and ischial rami or pubic symphysis was performed, depending on the extent of tumor invasion (Fig. 3-A and 3-B). Then, disarticulation of the sacroiliac joint was performed by gradually detaching the connected tissues using an electrotome. It should be noted that the sacroiliac joint is very wide and uneven, so it is necessary to dissect the ligamentous connection of the articular interface with an electrotome while gradually breaking it open.

The cartilage of the sacral articular interface was completely removed by curettage (Fig. 3-C). The subchondral bone was preserved to maintain the osseous strength (Fig. 3-D).

Because of the uneven surface of the sacroiliac joint, the prosthesis was seamlessly matched and installed (Fig. 3-E). The sacral tray of the prosthesis was placed to fully fit the ventral sacral interface to alleviate the shear force. Three to 5 screws were inserted into the sacrum to fix the prosthesis as planned (Table III). For 13 patients, ipsilateral lumbar pedicle screws were inserted at L5 for lumbosacral fixation to further enhance prosthetic stability (Fig. 3-E). Finally, the femoral stem prosthesis was routinely placed. The ligaments and muscles around the hip joint were reconstructed whenever possible.

Postoperative Management

The initiation of bed-based early rehabilitation started 2 weeks postoperatively. Full weight-bearing walking was not encouraged until 3 months after surgery. All 28 patients were followed for a minimum of 24 months or until death.

TABLE II Measurements of the Displacement Distance of the Hip Rotation Center and Postoperative Acetabular Angle

Patient	Displacement Distance* (mm)				Postoperative Acetabular Angle	
	Overall	Lateromedial	Anteroposterior	Craniocaudal	Inclination	Anteversion
1	6.23	0.11	0.71	6.18	40.93°	17.40°
2	17.30	12.76	8.54	7.96	57.81°	5.48°
3	9.07	4.83	7.19	2.65	37.35°	29.83°
4	19.67	18.71	0.02	6.08	45.04°	8.08°
5	32.41	28.33	8.67	13.13	45.79°	11.12°
6	24.86	9.83	0.91	22.81	66.76°	17.05°
7	32.08	11.72	14.82	25.92	59.18°	5.59°
8	20.29	4.24	16.61	10.83	52.82°	3.71°
9	16.92	16.30	3.00	3.36	41.35°	24.38°
10	19.39	0.81	18.61	5.38	54.00°	-4.99°
11	16.50	14.57	5.54	5.38	44.52°	0.21°
12	10.78	8.31	6.85	0.03	33.86°	-24.42°
13	5.47	2.10	3.37	3.75	45.51°	19.74°
14	25.33	5.05	4.38	24.42	44.61°	15.76°
15	6.43	4.00	4.94	0.90	32.28°	21.19°
16	7.13	6.28	1.11	3.17	36.39°	27.56°
17	22.38	16.54	14.71	3.28	33.65°	43.51°
18	12.95	5.86	10.42	4.96	42.57°	23.39°
19	9.09	4.64	7.02	3.43	37.72°	23.68°
20	3.25	1.64	1.86	2.09	39.32°	27.58°
21	3.70	1.80	2.02	2.52	41.82°	32.29°
22	7.75	5.23	2.16	5.28	39.54°	35.33°
23	6.32	3.86	2.09	4.54	39.28°	32.01°
24	3.44	3.21	1.20	0.23	38.49°	26.88°
25	14.38	2.05	2.04	14.08	44.10°	24.78°
26†						
27†						
28†						
Total (n = 25)†	14.12 ± 8.75	7.71 ± 6.86	5.95 ± 5.39	7.29 ± 7.34	43.79 ± 8.49°	17.89 ± 14.76°

*Data are given, and were analyzed, as absolute values. †No postoperative CT. ‡The values are given as the mean and standard deviation.

Assessment of Complications and Oncological and Functional Outcomes

Complications were assessed at the time of follow-up as scheduled. Aseptic loosening was defined as reported in a previous study¹². Using 3D modeling and matching techniques, as we previously reported¹³, the displacement distance of the hip rotation center (HRC) from before to after surgery and the postoperative inclination and anteversion were calculated. Oncological outcomes were assessed through a review of patient records. Lower-limb function was evaluated according to the Musculoskeletal Tumor Society (MSTS)-93 score at the latest follow-up¹¹.

Finite Element Analysis (FEA)

Details of the FEA are provided in the Appendix.

Statistical Analysis

Continuous variables were compared using an independent-samples t test or 1-way analysis of variance (ANOVA) for parametric data and a Mann-Whitney U test or Kruskal-Wallis test for nonparametric data. Categorical variables were compared with a Fisher exact test. Tests were 2-sided, and p values of <0.05 were considered significant. All statistical analyses were performed using IBM SPSS Statistics (version 22.0; IBM).

Results

In the overall cohort, the mean operative time (and standard deviation) was 6.2 ± 1.3 hours, and the mean blood loss was $1,443 \pm 466$ mL. The surgical margin was positive in 1 case; the

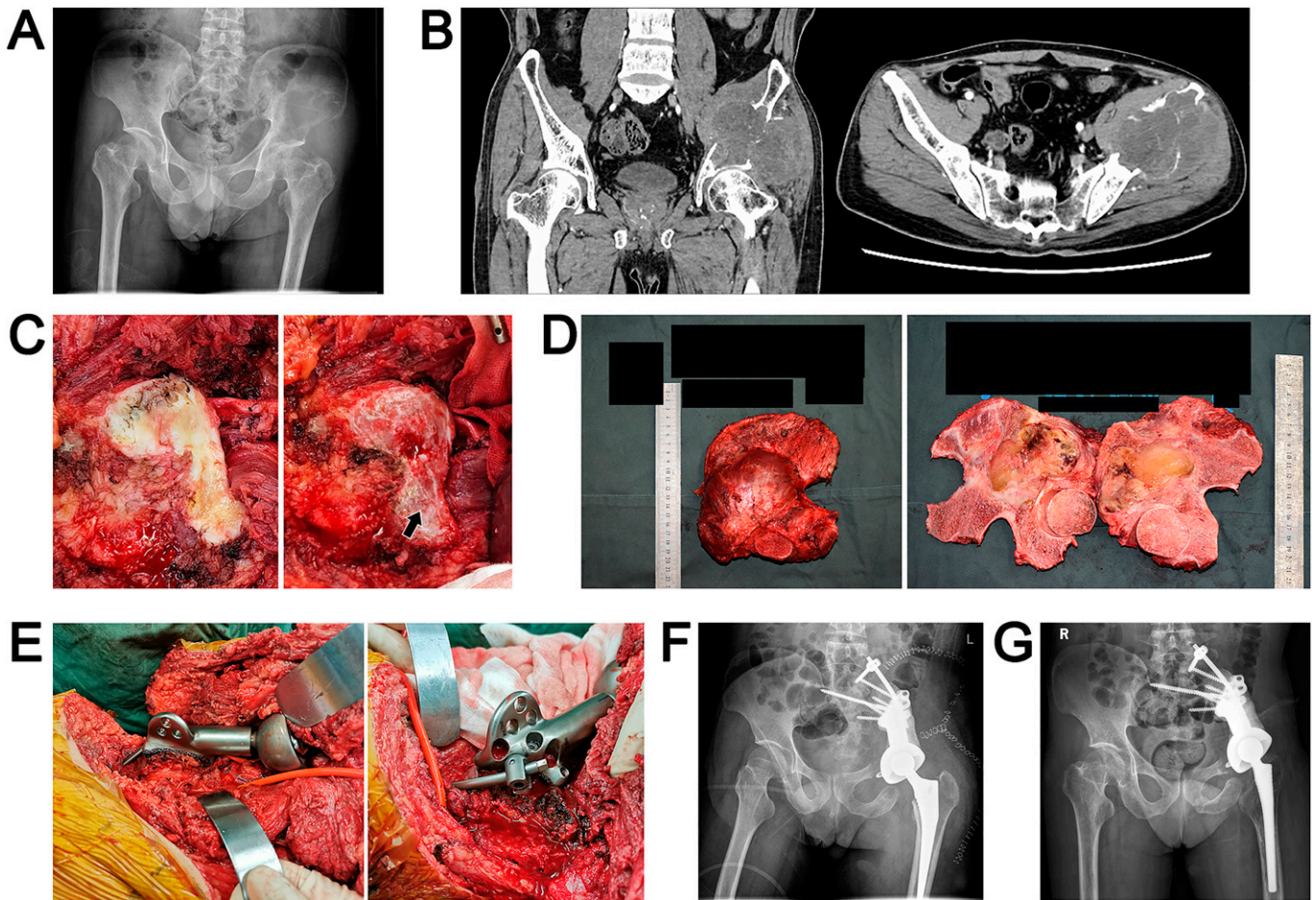


Fig. 3

A 44-year-old male patient with a malignant perivascular epithelioid cell tumor underwent prosthetic reconstruction using 5 sacral screws and lumbosacral fixation after type I + II internal hemipelvectomy. **Figs. 3-A and 3-B** A preoperative radiograph and CT images show an osteolytic tumor involving the ilium and acetabulum. **Fig. 3-C** Disarticulation of the sacroiliac joint was performed, and the articular cartilage of the sacral interface of the sacrum was removed (black arrow). **Fig. 3-D** En bloc resection of the tumor. **Fig. 3-E** The prosthesis was installed on the articular interface of the sacrum, and ipsilateral lumbar pedicle screws were inserted at L5 for lumbosacral fixation to further enhance prosthetic stability. **Fig. 3-F** A radiograph made immediately postoperatively shows that the prosthesis seamlessly matched the articular interface of the sacrum. **Fig. 3-G** A radiograph at the latest follow-up (24 months postoperatively) demonstrates no aseptic loosening, prosthetic breakage, or dislocation.

patient received radiation therapy and chemotherapy postoperatively. In the other 27 cases, a negative margin was achieved. Twelve patients received chemotherapy, 7 patients received immunotherapy, and 2 patients with thyroid cancer received iodine-131 treatment. No patients received radiation therapy postoperatively with the exception of the 1 patient with a positive margin as noted above.

Accuracy of Prosthesis Placement

For 24 of the 25 patients analyzed, all of the sacral screws were accurately inserted into the sacral bone as assessed on CT images postoperatively. Only 1 patient experienced a screw (at the S1 level) breaking the anterior cortical bone of the sacral canal. We speculated that this might have been because of small rotational errors in prosthesis installation. Fortunately, no neurological symptoms were observed.

The mean overall, lateromedial, anteroposterior, and craniocaudal displacement distances of the HRC from preoperatively to postoperatively were 14.12 ± 8.75 , 7.71 ± 6.86 , 5.95 ± 5.39 , and 7.29 ± 7.34 mm, respectively (Table II). The mean postoperative inclination and anteversion angles were $43.79^\circ \pm 8.49^\circ$ and $17.89^\circ \pm 14.76^\circ$, respectively.

Complications and Oncological and Functional Outcomes

At a mean follow-up of 36.2 months (range, 6.2 to 84.9 months), 17 patients were alive, including 11 who were free of disease (Table I). The rate of implant-related complications was 14.3% (4 of 28) for prosthetic breakage (Fig. 4), 14.3% (4 of 28) for aseptic loosening, and 7.1% (2 of 28) for dislocation (Table IV). The rate of superficial infection was 10.7% (3 of 28) and the rate of deep infection was 7.1% (2 of 28). For 1 of the patients with deep infection, the

TABLE III Comparisons of Demographic Data

	Lumbosacral Fixation			No. of Sacral Screws				Total (N = 28)
	Yes (N = 13)	No (N = 15)	P Value	3 (N = 10)	4 (N = 10)	5 (N = 8)	P Value	
Age* (yr)	53.3 ± 12.5	49.7 ± 14.9	0.492†	53.7 ± 13.2	54.1 ± 11.8	45.0 ± 15.9	0.308‡	51.4 ± 13.7
Sex (no.)			0.433§				0.785§	
Male	7	11		6	6	6		18
Female	6	4		4	4	2		10
Side (no.)			0.476§				0.802§	
Left	8	7		5	5	5		15
Right	5	8		5	5	3		13
Tumor region** (no.)			0.020§				0.126§	
I + II	9	3		7	3	2		12
I + II + III	4	12		3	7	6		16
Source of tumor (no.)			1.000§				0.308§	
Primary malignancy	10	12		6	9	7		22
Metastasis	3	3		4	1	1		6
Follow-up* (mo)	27.9 ± 6.8	43.5 ± 21.2	0.013#	41.7 ± 22.4	38.6 ± 17.4	26.4 ± 5.1	0.168‡	36.2 ± 17.8
No. of sacral screws (no.)			0.415§					
3	5	5						10
4	3	7						10
5	5	3						8
Lumbosacral fixation (no.)							0.415§	
Yes				5	3	5		13
No				5	7	3		15

*The values are given as the mean and standard deviation. †Independent-samples t test. ‡One-way analysis of variance (ANOVA). §Fisher exact test. #Mann-Whitney U test. **I + II = iliac and periacetabular areas, I + II + III = iliac, periacetabular, and pubic areas.

infection was resolved by means of debridement, while the other patient with deep infection underwent prosthesis removal.

The local recurrence and distant metastasis rates were 10.7% and 39.3%, respectively (Table IV). At the latest follow-up, the mean MSTS-93 score was 18.2 (range, 9 to 28).

Comparison of Groups with or without Lumbosacral Fixation

Comparisons between the groups with or without lumbosacral fixation are shown in Tables III and IV. The prosthetic breakage rate was lower for patients who underwent lumbosacral fixation (0 of 13, 0.0%) than for those who did not (4 of 15, 26.7%), although the difference did not reach significance ($p = 0.102$). No significant difference in the rate of aseptic loosening was identified between the groups with (1 of 13, 7.7%) or without (3 of 15, 20.0%) lumbosacral fixation.

Comparison of Groups with 3, 4, or 5 Sacral Screws

Comparisons among the groups with 3, 4, or 5 sacral screws are shown in Tables III and IV. The aseptic loosening rate was significantly greater in the group with 3 sacral screws (4 of 10, 40.0%) compared with the other 2 groups (4 screws, 0 of 10, 0.0%; and 5 screws, 0 of 8, 0.0%) ($p = 0.024$). In addition, inserting 3 sacral screws (3 of 10, 30.0%) resulted in a higher risk of prosthetic

breakage than use of 4 screws (1 of 10, 10.0%) or 5 screws (0 of 8, 0.0%), although no significant difference was detected ($p = 0.290$).

FEA of Stress

The distributions of the stresses on the postoperative pelvis, prosthesis, and screws were compared in the static standing position. The overall distributions of von Mises stresses relative to implant locations, sacral screws, and lumbosacral systems among the 5 models are shown in Figure 5.

The addition of lumbosacral fixation (Model 1 versus Model 2) visibly reduced the peak stress on the sacral screws by 48.2% (from 157.1 to 81.4 MPa) (Table V). Increasing the number of sacral screws from 3 to 4 (Model 1 versus Model 4) not surprisingly also alleviated the peak stress on the sacral screws by 28.4% (from 157.1 to 112.5 MPa). However, the peak stress decreased only by 11.5% (from 112.5 to 99.6 MPa) with a further increase from 4 to 5 screws (Model 4 versus Model 5). Halving all screw lengths exacerbated the peak stress on the sacral screws, from 157.1 to 233.2 MPa (Model 1 versus Model 3).

Discussion

Whether to reconstruct a stable pelvis after internal hemipelvectomy still remains controversial. Some studies^{14,15}

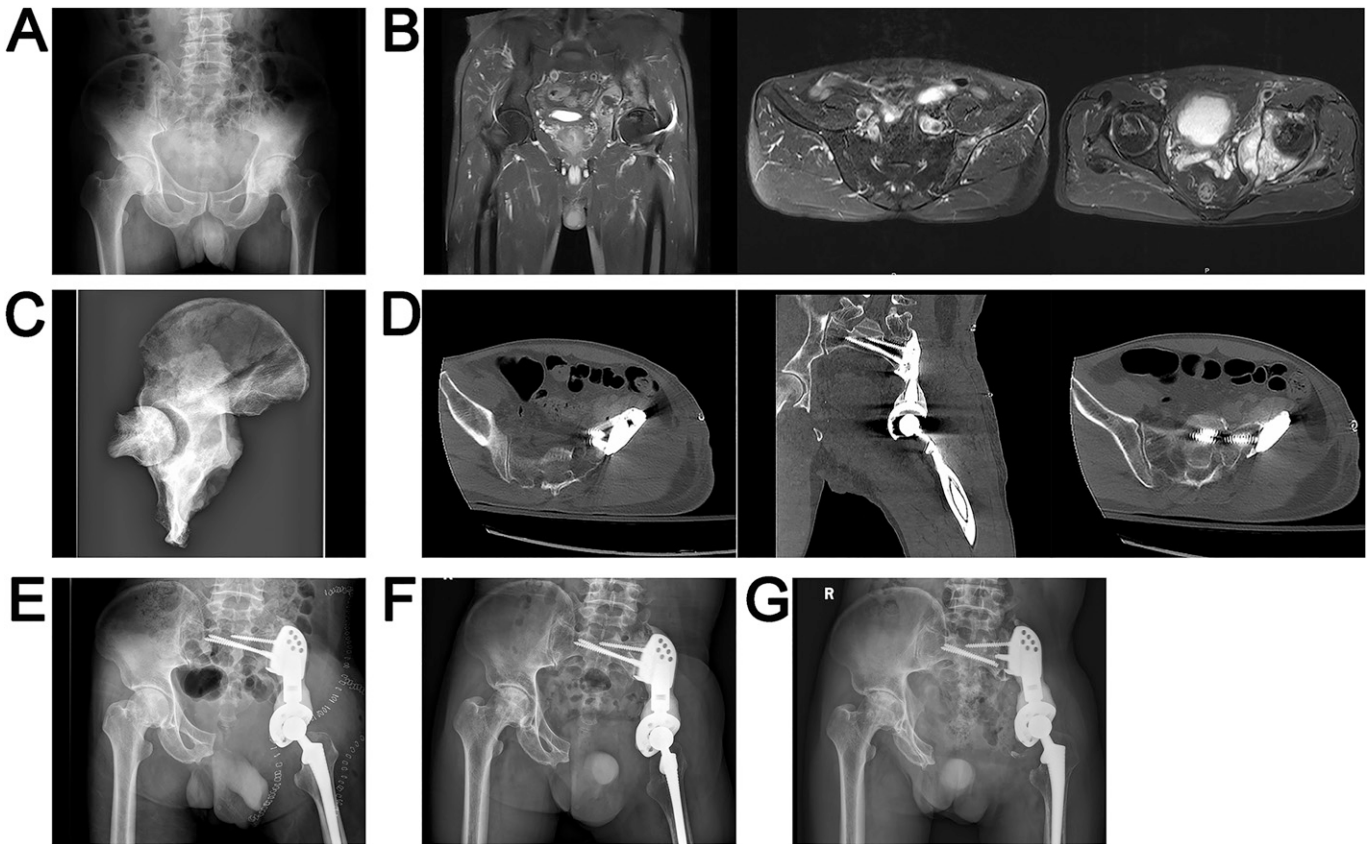


Fig. 4

A 59-year-old male patient with chondrosarcoma underwent prosthetic reconstruction using 3 sacral screws without lumbosacral fixation. **Figs. 4-A and 4-B** A preoperative radiograph and magnetic resonance imaging (MRI) scans (1.5-T contrast-enhanced T1-weighted images) show mixed osteogenic and osteolytic lesions involving the ilium, acetabulum, and pubis. **Fig. 4-C** Radiographic image of the tumor. **Fig. 4-D** Postoperative computed tomography (CT) images show that the prosthesis seamlessly matched the articular interface of the sacrum and that the sacral screws successfully avoided penetrating the sacral canal or anterior neurovascular plexus. Radiographs immediately postoperatively (**Fig. 4-E**), 12 months postoperatively (**Fig. 4-F**), and 21 months postoperatively show that sacral screw breakage occurred (**Fig. 4-G**).

TABLE IV Comparisons of Complications, Prognosis of Oncology, and Functional Outcomes*

	Lumbosacral Fixation			No. of Sacral Screws				Total (N = 28)†
	Yes (N = 13)*	No (N = 15)*	P Value†	3 (N = 10)*	4 (N = 10)*	5 (N = 8)*	P Value†	
Prosthetic breakage	0	4§	0.102	3	1	0	0.290	4 (14.3%)
Aseptic loosening	1	3	0.600	4	0	0	0.024	4 (14.3%)
Dislocation	1	1	1.000	0	1	1	0.735	2 (7.1%)
Infection	3	2	0.639	2	1	2	0.835	5 (17.9%)
Local recurrence	2	1	0.583	1	2	0	0.756	3 (10.7%)
Distant metastasis#	3	8	0.137	4	6	1	0.126	11 (39.3%)
MSTS-93 score	17.0 ± 5.3	19.3 ± 5.1	0.263**	16.2 ± 6.3	19.4 ± 3.8	19.3 ± 5.3	0.331††	18.2 ± 5.3

*The values are given as the number of patients, except for the MSTS-93 scores, which are given as the mean and standard deviation. †Fisher exact test, except where otherwise noted. ‡The values are given as the number of patients, with the percentage in parentheses, except for the MSTS-93 scores, which are given as the mean and standard deviation. §All 4 patients had breakage of screws. #Six patients experienced metastasis before surgery. **Independent-samples t test. ††One-way analysis of variance (ANOVA).

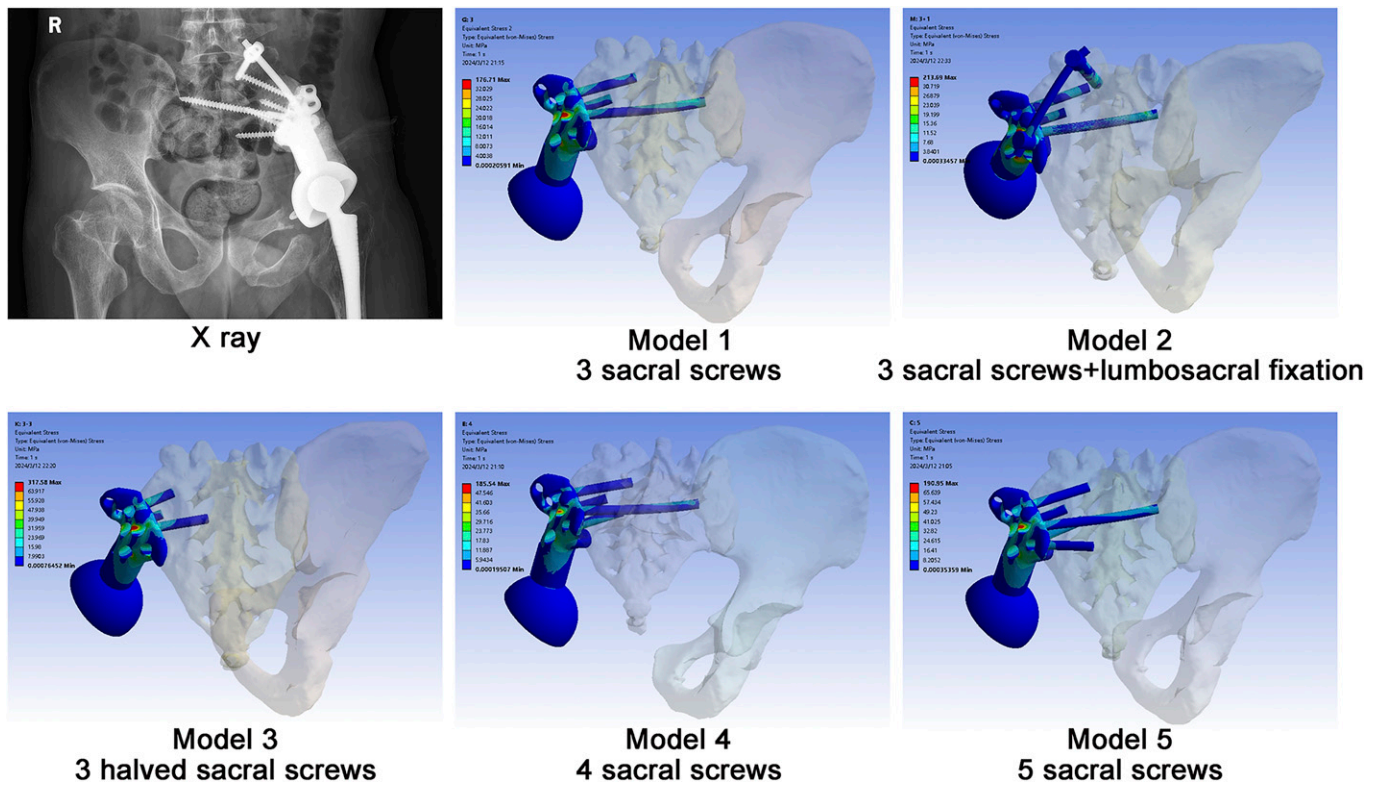


Fig. 5

Five typical models of von Mises stress constructed for comparison based on a single prototype (Patient Number 20 in Tables I and II) and the respective radiograph are shown: the initially designed prosthesis was fixed with 3 sacral screws (Model 1); 3 sacral screws and the lumbosacral system (Model 2); 3 sacral screws with the length of the screws halved (Model 3); 4 sacral screws (Model 4); and 5 sacral screws (Model 5).

have advocated for hip transposition, which showed comparable functional outcomes with lower rates of reoperation and infection than those of reconstructing the pelvis using prostheses, while avoiding the risk of implant failure. However, the possibilities of shortening of the leg, a change in gait, and a longer period of rehabilitation are also of concern¹⁶. Consequently, extensive research efforts have been dedicated to the construction of a stable pelvis after Enneking Type I + II (+ III) resection (Table VI). However, the reconstruction procedure is highly challenging after extensive type I + II (+ III) internal hemipelvectomy, since the prosthesis can only be fitted and fixed to the sacrum¹⁷. After an osteotomy using an ultra-

sonic bone knife or a line saw in most of the studies shown in Table VI, the predominantly flat sacral side of the bone-implant interface would be composed of loose cancellous bone and may be vulnerable to shear loading. Simple screw fixation can lead to screw breakage and prosthetic loosening. Moreover, the loss of the iliac wing makes it difficult to precisely relocate the acetabulum because of the complete removal of anatomical landmarks. In the present study, a novel 3D-printed, custom-made prosthesis with an articular interface was designed for accurate pelvic reconstruction after type I + II (+ III) tumor resection. The outcomes suggest that the HRC was accurately restored, that limb function was satisfactory for the majority of the patients, and

TABLE V Comparisons of Maximum and Average von Mises Stress on the 3D-Printed Prosthesis Under 700 N Preload

Model	No. of Sacral Screws	Lumbosacral Fixation	Stress on Sacral Screws (MPa)		Stress on Titanium Prosthesis (MPa)	
			Maximum	Average	Maximum	Average
Model 1	3	No	157.1	18.3	176.7	3.5
Model 2	3	Yes	81.4	13.0	213.7	4.0
Model 3	3 halved screws	No	233.2	30.8	317.6	6.6
Model 4	4	No	112.5	15.7	185.5	3.7
Model 5	5	No	99.6	13.0	191.0	6.1

TABLE VI Literature Review of Pelvic Reconstruction Approaches After Extensive Enneking Type I + II (+ III) Resection*

Study (Year)	Total No. of Cases	No. (%) of Enneking I + II (+ III) Resections	Prosthesis Used	Mean Follow-up (Range) (mo)	NED	AWD	DOD	Mean MSTS Score (%)†	PB	AL	Dislocation	WRC	LR	DM
Ji et al. (2013) ²²	100	16 (16%)	Modular hemipelvic prosthesis	52.9 (NM)	11	1	4	14.8 (49.2%)	NM	NM	NM	4	4	NM
Wafa et al. (2014) ²³	18	11 (61%)	Irradiated autografts	51.6 (4-185)	5	2	4	22.3 (74.2%)‡	1	1	2	4	0	3
Zhang et al. (2018) ¹⁸	20	5 (25%)	Modular hemipelvic prosthesis	39.4 (17-52)	4	0	1	17.4 (58.0%)	0	0	0	0	0	1
Guo et al. (2023) ¹⁹	4	4 (100%)	Custom-made prosthesis	16.5 (6-24)	3	1	0	19.5 (65%)	0	0	0	2	0	0
Kekeç et al. (2022) ²⁴	19	9 (47.4%)	Fresh-frozen pelvic allograft	66.6 (6-124)	6	0	3	20.6 (68.8%)	0	1	1	3	0	3
Hu et al. (2022) ²⁵	16	5 (31.3%)	Custom-made prosthesis	19.2 (9-39)	3	2	0	27.4 (91.3%)	0	0	0	2	NM	NM
Peng et al. (2020) ²⁶	6	5 (83.3%)	Custom-made prosthesis	11.0 (5-24)	5	0	0	19.8 (66%)§	NM	0	0	0	0	0
Xie et al. (2022) ²⁷	14	8 (57.1%)	Acetabular component with sacral components	32.0 (20-62)§	7	1	0	22.9 (76.3%)	0	0	0	2	0	1
Tang et al. (2017) ²⁸	13	8 (61.5%)	Acetabular component with pedicle screw-rod system	27.0 (8-43)	4	4	0	23.7 (78.9%)	0	0	1	1	1	0
Escudero-Acurio et al. (2022) ²⁹	1	1 (100%)	Custom-made hydroxyapatite Schanz prosthesis	10	1	0	0	NM	0	0	0	0	0	0
Beadel et al. (2005) ³⁰	26	21 (80.8%)	Fresh-frozen irradiated hemipelvic allografts#	NM	8	3	10	13.5 (45%)**	NM	2	5	9	3	3
Çolak et al. (2020) ³¹	1	1 (100%)	Pedestal conic cup	12	1	0	0	29 (96.7%)	0	1††	0	0	0	0
Wang et al. (2020) ³²	13	13 (100%)	Custom-made prosthesis	27.3 (25-31)	11	2	0	23 (76.7%)‡‡	0	0	0	2	0	1
Wang et al. (2018) ³³	11	Unknown	Custom-made prosthesis	15.5 (NM)	9	1	1	19.2 (64%)	0	0	2	1	0	1
Guo et al. (2010) ³⁴	18	6 (33.3%)	Modular hemipelvic prosthesis	41 (7-73)	10	3	5	21.5 (71.7%)	0	1	2	2	4	6
Qu et al. (2019) ³⁵	22	22 (100%)	A femoral head autograft plus a modular hemipelvic prosthesis	47.2 (15-125)	14	1	7	18.2 (60.8%)	0	0	2	3	5	8
Hoffmann et al. (2006) ¹⁵	45	18 (40%)	Hip transposition	NM	NM	NM	NM	18.2 (60.8%)§§	NA	NA	NA	NM	NM	NM
Kunisada et al. (2019) ¹⁴	7	5 (71.4%)	Hip transposition	57.2 (18-102)	0	3	2	20.2 (67.3%)	NA	NA	NA	1§	1	4§
Zhang et al. (2023) ¹⁶	38	11 (28.9%)	Hip transposition	17‡‡ (6-110)§	23§	6§	9§	15.9 (53%)	NA	NA	NA	16§	0##	13§
Current study	28	28 (100%)	Custom-made prosthesis	36.2 (6.2-84.9)	11	6	11	18.2 (60.8%)	4	4	2	5	3	11

*The data in the table are limited to the patients undergoing Enneking I + II (+ III) resections unless otherwise specified. NED = no evidence of disease, AWD = alive with disease, DOD = died of disease, PB = prosthetic breakage, AL = aseptic loosening, WRC = wound-related complications, LR = local recurrence, DM = distant metastasis, NM = not mentioned, and NA = not applicable. †Percentages are of a total possible score of 30. ‡Data of 1 patient were missing. §The data were calculated on the basis of all patients in the study. #One patient had a fresh-unfrozen osteochondral allograft. **The data of only 6 patients with a minimum follow-up of 60 months were analyzed. ††One patient had implant failure resulting from a motorcycle accident. ‡‡The data are presented as the median instead of the mean. §§Two unrelated patients were also included and analyzed. ##Patients with local recurrence were excluded from the study.

that, overall, relatively few prosthesis-related complications were found.

In this study, we aimed to establish a safe osteotomy margin, accurately restore the HRC, and achieve low mechanical failure via our customized prosthesis after type I + II (+ III) internal hemipelvectomy. The use of this prosthesis has 3 advantages. First, the 3D-printed articular interface of the prosthesis was designed to seamlessly match the irregular sacral interface. Thus, the position of the prosthesis could be uniquely determined during installation, allowing placement of the acetabular cup at the native position. The mean anteroposterior and craniocaudal displacement distances of the HRC in the present study were 5.95 and 7.29 mm, respectively, which were fully within the safe range of dislocation risk (<18 mm and <20 mm, respectively), as reported in our previous study¹³. In addition, an acceptable dislocation rate (7.1%, 2 of 28) was identified for the cohort, which appears to be favorable compared with the rates of most other types of reconstructions (Table VI). Second, the relatively large, seamless, and porous bone-prosthesis interface, the novel sacral tray structure, and the lumbosacral fixation system collectively relieved shear force and may promote osseointegration, subsequently reducing implant-related complications and leading to stable reconstruction. In this study, although the total rate of the prosthesis-related complications of aseptic loosening, prosthetic breakage, and dislocation reached 28.6% (8 of 28), this rate decreased for patients who underwent lumbosacral fixation (15.4%, 2 of 13) or who had 4 or 5 sacral screws implanted (16.7%, 3 of 18). Thus, short to medium-term mechanical complications may be able to be controlled during the process of prosthesis design and fixation. In addition, the rate of the prosthesis-related complication of deep infection was 7.1% (2 of 28), which is no higher than that reported for hip transposition^{14,16}. The rapid and relatively straightforward installation of the prosthesis could reduce the operative time and thus might reduce the infection rate. As for the porous “bone-implant” interface, Zhang et al.¹⁸ reported on a series of hemipelvic reconstructions using a prosthesis with a 3D-printed porous interface and showed the occurrence of osseointegration at the bone-prosthesis interface as assessed by hard-tissue slicing and micro-CT scanning, with 1 case requiring a second external hemipelvectomy for local recurrence. However, the evidence of osseointegration could not be captured in our cases since we have not performed a revision surgery, in which we can check the osseointegration of the bone-implant interface by pathology. Third, the number and trajectories of sacral screws were planned on an individualized basis to avoid injuring the surrounding neurovascular structures. A high degree of safety and convenience of operation were achieved during the prosthesis fixation process. Our newly designed prosthesis demonstrated acceptable early functional outcomes, which seemed no inferior to those reported by other studies (Table VI).

Good biomechanical properties are essential for the design of artificial implants to prevent prosthesis-related complications¹⁹. In the present study, we observed that the presence of lumbosacral fixation or an increase in the number of sacral screws from 3 to 4 or 5 decreased the risk of instrumentation failure. This result was further confirmed by the finite element analysis, which showed that

lumbosacral fixation or more sacral screws could decrease and disperse the shear force exerted on each screw. This vertical force would gradually contribute to the breakage of screws and prosthetic loosening. Similar phenomena have also been confirmed by multiple studies^{19,21}. On the basis of these findings, we recommend the use of 4 to 5 sacral screws to improve the stability of prostheses in clinical practice. If there is only room to accommodate 3 sacral screws, the use of a unilateral lumbosacral prosthesis fixation system is highly recommended. In addition, halving all screw lengths exacerbates the peak stress on the sacral screws. Thus, ensuring sufficient sacral screw length is necessary for a stable fixation.

This study had a number of limitations. First, it was a retrospective study without a control group. We cannot directly compare the outcomes with available alternatives. Second, a minimum of 2 years of follow-up is insufficient to fully assess prosthesis durability, and we will continue to follow this cohort. Third, the finite element model for the biomechanical analysis was simplified to reduce computational complexity. Only static-load analysis was incorporated into the study; gait analysis with a motion capture system should be the focus of a future study. Fourth, the follow-up duration of the lumbosacral fixation group was shorter than that of the group without lumbosacral fixation. Initially, the prosthesis was designed only for fixation with sacral screws. However, as relatively high aseptic loosening and prosthetic breakage rates were observed, especially when only 3 sacral screws were used for fixation, the lumbosacral system was introduced and used when possible, with more stable fixation expected. This development of the fixation approach contributed to the difference in follow-up duration, which might result in bias in the mechanical failure analysis.

In summary, the use of a 3D-printed hemipelvic prosthesis could help surgeons reconstruct the HRC at the native position and lead to satisfactory functional recovery. The lumbosacral fixation system and more sacral screws combined with the sacral tray system worked efficiently to alleviate shear force, which may decrease the risk of prosthesis-related complications. The seamlessly matched and porous bone-prosthesis interface might achieve osseointegration and long-term stabilization.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at <http://links.lww.com/JBJS/I230>. ■

NOTE: The authors thank Junyan Xie, Jiadan Wu, and Keyi Wang from the Second Affiliated Hospital of Zhejiang University for their assistance in collecting data.

Xin Huang, MD^{1,2,3,4}
Donghua Huang, MD^{1,2,3,4}
Nong Lin, MD^{1,2,3,4}
Xiaobo Yan, MD^{1,2,3,4}
Hao Qu, MD^{1,2,3,4}
Zhaoming Ye, PhD^{1,2,3,4}

¹Musculoskeletal Tumor Center, Department of Orthopedics, The Second Affiliated Hospital of Zhejiang University School of Medicine, Hangzhou, Zhejiang, People's Republic of China

²Orthopedics Research Institute of Zhejiang University, Hangzhou, Zhejiang, People's Republic of China

³Key Laboratory of Motor System Disease Research and Precision Therapy of Zhejiang Province, Hangzhou, Zhejiang, People's Republic of China

⁴Clinical Research Center of Motor System Disease of Zhejiang Province, Hangzhou, Zhejiang, People's Republic of China

Email for corresponding author: yezhaoming@zju.edu.cn

References

- Dang J, Fu J, Liu D, Zhang Z, Mi Z, Cheng D, Liu X, Zhang Y, Zhu D, Wang L, Shi Y, Fan H. Clinical application of 3D-printed patient-specific guide plate combined with computer navigation in acetabular reconstruction following resection of peri-acetabular tumors. *Ann Transl Med.* 2022 Jan;10(2):76.
- Hipfl C, Stihsen C, Puchner SE, Kaider A, Dominkus M, Funovics PT, Windhager R. Pelvic reconstruction following resection of malignant bone tumours using a stemmed acetabular pedestal cup. *Bone Joint J.* 2017 Jun;99-B(6):841-8.
- Witte D, Bernd L, Bruns J, Goshager G, Harges J, Hartwig E, Lehner B, Melcher I, Mutschler W, Schulte M, Tunn PU, Wozniak W, Zehlten-Hinguranage A, Zeifang F. Limb-salvage reconstruction with MUTARS hemipelvic endoprosthesis: a prospective multicenter study. *Eur J Surg Oncol.* 2009 Dec;35(12):1318-25.
- Wu J, Xie K, Luo D, Wang L, Wu W, Yan M, Ai S, Dai K, Hao Y. Three-dimensional printing-based personalized limb salvage and reconstruction treatment of pelvic tumors. *J Surg Oncol.* 2021 Sep;124(3):420-30.
- Park JW, Kang HG, Kim JH, Kim HS. The application of 3D-printing technology in pelvic bone tumor surgery. *J Orthop Sci.* 2021 Mar;26(2):276-83.
- Feng Q, Li Z, Zhang X, Feng J, Wang D. Study of the efficacy of 3D-printed prosthetic reconstruction after pelvic tumour resection. *J Biomater Appl.* 2023 Apr;37(9):1626-31.
- Guo W, Sun X, Ji T, Tang X. Outcome of surgical treatment of pelvic osteosarcoma. *J Surg Oncol.* 2012 Sep 15;106(4):406-10.
- Zhou Y, Duan H, Liu Y, Min L, Kong Q, Tu C. Outcome after pelvic sarcoma resection and reconstruction with a modular hemipelvic prostheses. *Int Orthop.* 2011 Dec;35(12):1839-46.
- Guo W, Li D, Tang X, Ji T. Surgical treatment of pelvic chondrosarcoma involving periacetabulum. *J Surg Oncol.* 2010 Feb 1;101(2):160-5.
- Xiong Y, Lang Y, Tu C, Duan H. Pelvis metastasis from primary choroidal melanoma: a case report. *Oncotargets Ther.* 2014 Nov 14;7:2107-10.
- Enneking WF, Dunham W, Gebhardt MC, Malawar M, Pritchard DJ. A system for the functional evaluation of reconstructive procedures after surgical treatment of tumors of the musculoskeletal system. *Clin Orthop Relat Res.* 1993 Jan;(286):241-6.
- Deng Y, Smith PN, Li RW. Diabetes mellitus is a potential risk factor for aseptic loosening around hip and knee arthroplasty. *BMC Musculoskelet Disord.* 2023 Apr 5; 24(1):266.
- Qu H, Mou H, Wang K, Tao H, Huang X, Yan X, Lin N, Ye Z. Risk factor investigation for hip dislocation after periacetabular tumour resection and endoprosthetic reconstruction via thin-slice CT-based 3D model. *Bone Joint J.* 2022 Oct;104-B(10):1180-8.
- Kunisada T, Fujiwara T, Hasei J, Nakata E, Senda M, Ozaki T. Temporary External Fixation Can Stabilize Hip Transposition Arthroplasty After Resection of Malignant Periacetabular Bone Tumors. *Clin Orthop Relat Res.* 2019 Aug;477(8):1892-901.
- Hoffmann C, Goshager G, Gebert C, Jürgens H, Winkelmann W. Functional results and quality of life after treatment of pelvic sarcomas involving the acetabulum. *J Bone Joint Surg Am.* 2006 Mar;88(3):575-82.
- Zhang L, Iwata S, Saito M, Nakagawa M, Tsukushi S, Yoshida S, Gokita T, Ae K, Nakashima S, Watanuki M, Akiyama T. Hip Transposition Can Provide Early Walking Function After Periacetabular Tumor Resection: A Multicenter Study. *Clin Orthop Relat Res.* 2023 Dec 1;481(12):2406-16.
- Wang J, Min L, Lu M, Zhang Y, Wang Y, Luo Y, Zhou Y, Duan H, Tu C. Three-dimensional-printed custom-made hemipelvic endoprosthesis for primary malignancies involving acetabulum: the design solution and surgical techniques. *J Orthop Surg Res.* 2019 Nov 27;14(1):389.
- Zhang Y, Tang X, Ji T, Yan T, Yang R, Yang Y, Wei R, Liang H, Guo W. Is a Modular Pedicle-hemipelvic Endoprosthesis Durable at Short Term in Patients Undergoing Enneking Type I + II Tumor Resections With or Without Sacroiliac Involvement? *Clin Orthop Relat Res.* 2018 Sep;476(9):1751-61.
- Guo Z, Peng Y, Shen Q, Li J, He P, Yuan P, Liu Y, Que Y, Guo W, Hu Y, Xu S. Reconstruction with 3D-printed prostheses after type I + II + III internal hemipelvectomy: Finite element analysis and preliminary outcomes. *Front Bioeng Biotechnol.* 2023 Jan 9;10:1036882.
- Ji T, Yang Y, Tang X, Liang H, Yan T, Yang R, Guo W. 3D-Printed Modular Hemipelvic Endoprosthetic Reconstruction Following Periacetabular Tumor Resection: Early Results of 80 Consecutive Cases. *J Bone Joint Surg Am.* 2020 Sep 2; 102(17):1530-41.
- Wang B, Sun P, Xie X, Wu W, Tu J, Ouyang J, Shen J. A novel combined hemipelvic endoprosthesis for peri-acetabular tumours involving sacroiliac joint: a finite element study. *Int Orthop.* 2015 Nov;39(11):2253-9.
- Ji T, Guo W, Yang RL, Tang XD, Wang YF. Modular hemipelvic endoprosthesis reconstruction—experience in 100 patients with mid-term follow-up results. *Eur J Surg Oncol.* 2013 Jan;39(1):53-60.
- Wafa H, Grimer RJ, Jeys L, Abudu AT, Carter SR, Tillman RM. The use of extracorporeally irradiated autografts in pelvic reconstruction following tumour resection. *Bone Joint J.* 2014 Oct;96-B(10):1404-10.
- Kekeç AF, Güngör BŞ. Mid-term outcomes of hemipelvic allograft reconstruction after pelvic bone tumor resections. *Jt Dis Relat Surg.* 2022;33(1):117-31.
- Hu X, Chen Y, Cai W, Cheng M, Yan W, Huang W. Computer-Aided Design and 3D Printing of Hemipelvic Endoprosthesis for Personalized Limb-Salvage Reconstruction after Periacetabular Tumor Resection. *Bioengineering (Basel).* 2022 Aug 18; 9(8):400.
- Peng W, Zheng R, Wang H, Huang X. Reconstruction of Bony Defects after Tumor Resection with 3D-Printed Anatomically Conforming Pelvic Prostheses through a Novel Treatment Strategy. *Biomed Res Int.* 2020 Dec 1;2020:8513070.
- Xie X, Jin Q, Zhao Z, Wang Y, Wang B, Zou C, Yin J, Huang G, Shen J. A novel limb-salvage reconstruction strategy with a custom hemipelvic endoprosthesis and preserved femoral head following the resection of periacetabular tumors: A preliminary study. *J Surg Oncol.* 2022 Sep;126(4):804-13.
- Tang X, Guo W, Yang R, Yan T, Tang S, Li D. Acetabular Reconstruction With Femoral Head Autograft After Intraarticular Resection of Periacetabular Tumors is Durable at Short-term Followup. *Clin Orthop Relat Res.* 2017 Dec;475(12):3060-70.
- Escudero-Acurio P, Mahaluf F, Bahamonde L. Reconstruction of Type I-II Internal Hemipelvectomy in a Patient With Pelvic Myxoid Chondrosarcoma: A Case Report. *Cureus.* 2022 Jul 6;14(7):e26621.
- Beadel GP, McLaughlin CE, Wunder JS, Griffin AM, Ferguson PC, Bell RS. Outcome in two groups of patients with allograft-prosthetic reconstruction of pelvic tumor defects. *Clin Orthop Relat Res.* 2005 Sep;438(438):30-5.
- Çolak TS, Ahmet Fevzi Kekeç AFKAFK. Adolescent pelvic chondrosarcoma, surgical treatment, and unusual reconstruction with pedestal conic cup (LUMIC®): A case report. *Jt Dis Relat Surg.* 2020;31(3):648-52.
- Wang J, Min L, Lu M, Zhang Y, Wang Y, Luo Y, Zhou Y, Duan H, Tu C. What are the Complications of Three-dimensionally Printed, Custom-made, Integrative Hemipelvic Endoprostheses in Patients with Primary Malignancies Involving the Acetabulum, and What is the Function of These Patients? *Clin Orthop Relat Res.* 2020 Nov; 478(11):2487-501.
- Wang B, Hao Y, Pu F, Jiang W, Shao Z. Computer-aided designed, three dimensional-printed hemipelvic prosthesis for peri-acetabular malignant bone tumour. *Int Orthop.* 2018 Mar;42(3):687-94.
- Guo Z, Li J, Pei GX, Li XD, Wang Z. Pelvic reconstruction with a combined hemipelvic prostheses after resection of primary malignant tumor. *Surg Oncol.* 2010 Jun;19(2):95-105.
- Qu H, Li D, Tang S, Zang J, Wang Y, Guo W. Pelvic reconstruction following resection of tumour involving the whole ilium and acetabulum. *J Bone Oncol.* 2019 Mar 27;16:100234.